

ASCII MASTER FLOW – PANEL 1/12: Energy-service value rail

PRIMARY RESOURCE / IMPORT

-> CONVERSION + NETWORK

-> DISTRIBUTION + END USE

-> RELIABLE AFFORDABLE SERVICE

MUST REMEMBER: Energy infrastructure connects primary energy, conversion, generation,...

ASCII MASTER FLOW – PANEL 2/12: Power-energy distinction

MW / GW -> power capacity

MWh / GWh -> energy flow

CAPACITY SHARE != generation share

ELECTRICITY != total energy

ASCII MASTER FLOW – PANEL 3/12: Electricity-layer map

GENERATION

TRANSMISSION + SYSTEM OPERATION

DISTRIBUTION + RETAIL

MARKET + REGULATION

ASCII MASTER FLOW – PANEL 4/12: Real-time balance board

GENERATION + IMPORT + DISCHARGE

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DEMAND + EXPORT + CHARGE + LOSS

RESERVE + FLEXIBILITY protect balance

ASCII MASTER FLOW – PANEL 5/12: Institutional perimeter

CEA -> technical planning
CERC / SERC -> regulation
GRID-INDIA / SLDC -> operation
DISCOM -> retail service

ASCII MASTER FLOW – PANEL 6/12: Tariff-finance split

EXPLICIT STATE SUBSIDY

CROSS-SUBSIDY

REGULATORY ASSET

DISCOM DEBT / UNDER-RECOVERY

ASCII MASTER FLOW – PANEL 7/12: DISCOM diagnostic

AT&C -> energy recovery

ACS-ARR -> revenue-cost gap

PPA + SUBSIDY + COLLECTION

ONE RATIO cannot diagnose all stress

ASCII MASTER FLOW – PANEL 8/12: Petroleum perimeter

UPSTREAM -> production

MIDSTREAM -> pipeline + storage

DOWNSTREAM -> refining + marketing

PNGRB excludes crude + gas production

ASCII MASTER FLOW – PANEL 9/12: Energy-security dimensions and resilience pentagon

PHYSICAL AVAILABILITY

INFRASTRUCTURE ACCESSIBILITY

USER AFFORDABILITY

SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE

ASCII MASTER FLOW – PANEL 10/12: Renewable-integration chain

VARIABLE CAPACITY

-> GRID + FORECAST

-> FLEXIBILITY + STORAGE

-> RELIABLE GENERATION VALUE

CLOSE DISTINCTION: Capacity in MW is not generation in MWh, plant load factor is not...

ASCII MASTER FLOW – PANEL 11/12: Status and unit ladder

ANNOUNCED / BILL

APPROVED / SANCTIONED / OUTLAY

INSTALLED CAPACITY

GENERATION + SERVICE OUTCOME

EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State fuel/technology, stock or flow, unit,...

ASCII MASTER FLOW – PANEL 12/12: Energy-economics answer spine

DEFINE carrier + unit + layer

TRACE network + tariff + regulator

TEST security + transition

CONCLUDE service, not headline capacity